

WASP-43b

R-2726

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_120409 · created 2026-09-01T23:37:55+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456026.71428 +/- 0.00037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.041
Transit depth (Rp/Rs)^2	1.82 +/- 1.1 [%]
Orbital Inclination (inc)	84.1 +/- 2.59
Airmass coefficient 1 (a1)	0.887 +/- 0.022
Airmass coefficient 2 (a2)	0.083 +/- 0.043
Scatter in the residuals of the lightcurve fit is	2.5 %
Best Comparison Star	#2 - [549, 372]
Optimal Aperture	2.59
Optimal Annulus	7.26
Transit Duration (day)	0.0533 +/- 0.007

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.135 ± 0.041 vs 0.1594 (deviation 0.59σ)
Duration fit vs published	0.0533 d vs 0.0512 d (deviation 0.29σ)
Mid-time O–C	-0.8 min
Depth SNR	3.3
Residual scatter	2.5 %

Light curve

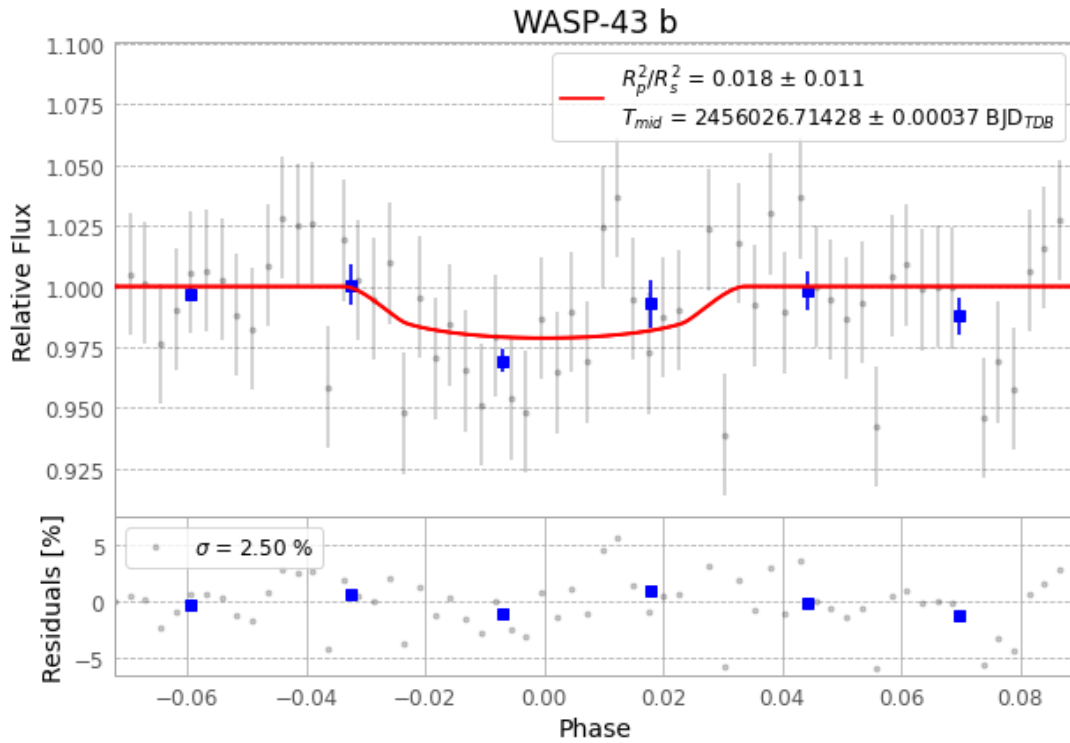


Figure 1 — FinalLightCurve_WASP-43 b_2012-04-08.png (SHA-256 80ec2940896cf1c8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [242, 293], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7f428f8c98f8a0fb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), WASP-43120409033615.FITS → WASP-43120409064513.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-120409.log (b5e59c20ae95...), FinalLightCurve_WASP-43 b_2012-04-08.png (80ec2940896c...), FinalLightCurve_WASP-43 b_2012-04-08.csv (3ecd857e083...)

Compute resources

Wall time 259.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)